

SECTION A (50 MARKS)

INSTRUCTION: ANSWER ALL QUESTIONS ON THE QUESTION PAPER

CIRCLE ALL THE CORRECT ANSWERS

1. Factors that control weight include all these except
A. dieting B. Peer influence C. innovations. D. culture
2. Fruits, vegetables and cereals are potent sources of:
A. Antioxidants B. Unsaturated fat C. Saturated fat D. Free radicals
3. The leading source of antioxidants in the U.S. diet is:
A. Citrus fruits B. Spinach C. Coffee C. Egg yolks
4. The essential fatty acids that must be derived from the diet are:
A. Stearidonic acid and eicosatetraenoic acid
B. Eicosapentaenoic acid and docosapentaenoic acid
C. Linoleic and alpha-linoleic acid
D. Gamma-linoleic acid and arachidonic acid
5. All of the following statements about omega-3 fatty acids are true except:
A. They help to maintain healthy triglyceride and high-density lipoprotein
B. They have significantly contributed to the obesity epidemic
C. They are necessary for healthy infant growth and development
D. They play an important role in the production of hormones that govern numerous metabolic and biological processes
6. All of the following may be associated with scurvy except:
A. Loss of appetite and irritability
B. Diarrhoea and fever
C. Tenderness and swelling in legs

D. First symptom is altered mental status

7. The only fat-soluble antioxidant synthesized in the body is?

A. Vitamin D B. Thiamine C. Ascorbic acid D. CoQ10

8. Good source of vitamin D include all except:

A. blueberries B. Sunlight C. Salmon, tuna sardines and mackerel D. Fortified milk and other dairy products

9. One of the fat-soluble vitamins involved in coagulation is:

A. Vitamin K B. Vitamin A C. Vitamin D D. Vitamin E

10. Products that contain live microorganisms in sufficient numbers to alter intestinal microflora and promote intestinal microbial balance are known as:

A. Antibiotics B. Probiotics C. Fruits and vegetables D. Digestive enzymes

11. Nondigestible food ingredients that stimulate the growth and activity of certain bacteria in the colon are called:

A. Insoluble fibre B. Probiotics C. Prebiotics D. Cellulose

12. A deficiency of thiamine (vitamin B1) in the diet causes:

A. Osteopenia B. Beri-beri C. Protein malnutrition D. Scurvy

13. The USDA Dietary Guidelines for Americans advise:

A. Limiting carbohydrates to 10 percent of daily calories
B. Limiting total fat intake to 20 to 35 percent of calories
C. Limiting protein to 10 percent of daily calories

D. Limiting intake of fats and oils to 10 percent of daily calories

14. All of the following statements about vitamin B3 (niacin) are true except:

A. It helps to release energy in carbohydrates, fat, and protein

B. It improves blood lipid levels

C. Deficiency causes beriberi

D. It is involved in the synthesis of sex hormones

15. The average American consumes approximately how much sodium per day?

A. 2,300 mg B. 230 mg C. 340 mg D. 3,400 mg

16. All of the following are potentially modifiable risk factors for osteoporosis except:

A. Anorexia nervosa

B. Chronically low intake of calcium and vitamin D

C. Chronically low intake of vitamins C and B6

D. Excessive alcohol consumption

17. Consuming fewer than 40grams of proteins per day may lead to:

A. Hypoglycemia B. Kwashiorkor C. Marasmus D. Ketosis

18. Characteristics of successful dieters include all of the following except:

A. Maintaining a daily food journal

B. Counting calories

C. Adhering to a strict eating plan

D. Eliminating all carbohydrates from their diets

19. Iron supplements are frequently recommended for all of the following except:

A. Women who are pregnant

B. Sedentary workers

C. Teenage girls

D. Post-menopausal women

20. Ancient people saw food to have Powers

A. healing. B. physical C. emotional. D. psychological.

21. In the history of nutrition what was the vitamin deficiency British soldiers suffered from

A. B B. B C. C D. D

22. Which of these foods aids in controlling osteoporosis?

A. Milk B. Kontomire C. orange D. sunlight

23. The science of how the body uses food is

A. Nutrients B. Nutrition C. Food science D. Dietetics

24. The fuel for cell activity is known as

A. Nutrients B. Nutrition C. Food science D. Dietetics

25. Chocolate is an example of

A. nutrient dense food B. energy dense food C. vitamin dense food D. empty calorie food

26. Water is an example of

A. nutrient dense food B. energy dense food C. vitamin dense food D. empty calorie food

27. Water makes up about ----- of the body

A. 50%. B. 70%. C. 60%. D. 80%.

28. An example of inorganic nutrient is

A. vitamin C. B. Vitamin D. C. protein. D. minerals.

29. An example of organic nutrient is
A. vitamin C. B. Vitamin D. C. protein. D. minerals.
30. Which of these is an example of macronutrient?
A. vitamin C. B. Vitamin D. C. protein. D. minerals.
31. The following are all carbohydrates except
A. Starch. B. sugar. C. butter. D. fibre.
32. Which of these is a secondary source of energy for the brain
A. Carbohydrates B. Protein. C. fats. D. vitamins.
33. Which of these is a primary source of energy for the body
A. Carbohydrates B. Protein. C. fats. D. vitamins.
34. Fat soluble vitamins include the following except
A. A B. E C. C D. D
35. The following fall under saturated fats except
A. palm kernel oil B. palm oil C. cocoa butter D. olive oil
36. What makes protein ideal in all food nutrients?
A. Carbon B. nitrogen C. oxygen D. carbon-dioxide
37. Pernicious anaemia is as a result of lack of vitamin B
A. 6 B. 12 C. 3 D. 2
38. Pellagra is as a result of lack of vitamin B
A. 6 B. 12 C. 3 D. 2

39. A set of values that serve as standards for nutrient intakes for healthy persons.

A. DRI B. UL C. EAR D. RDA

40. Intake value sufficient to meet nutrient requirement of 97-98% of all healthy individuals in a group is referred to as

A. DRI B. UL C. EAR D. RDA

41. Maximum intake level above which toxicity would increase is indicated by

A. DRI B. UL C. EAR D. RDA

42. Intake value estimated to meet requirement of half the healthy individuals in a group is

A. DRI B. UL C. EAR D. RDA

43. The passage process of digested nutrients through the walls of the intestines or stomach into the blood or lymph, where they are transported to the cells is

A. Digestion B. Secretion C. absorption D. Metabolism

44. All the chemical processes by which nutrients are used to support life, including anabolism and catabolism is

A. Digestion B. Secretion C. absorption D. Metabolism

45. This is the ability to acquire and use energy from the environment

A. Digestion B. Secretion C. absorption D. Metabolism

46. Basal conditions include the following except

A. not eaten food for at least 12 hours

B. a night of restful sleep

C. no strenuous activity is performed for at least 1 hour before the test

D. Being tickled by your lover

47. Breakdown of large molecules into simple ones to produce energy is referred to as

A. Catabolism B. Secretion C. absorption D. Metabolism

48. Food hygiene practices include the following except

A. The consumption of food and influencing factors

B. The prevention of food poisoning

C. Personal hygiene

D. Cleaning the Kitchen after use

49. The practices and associated attitudes that will determine what, when, why and how an individual will eat when alternatives are available to make a choice is his or her

A. food pattern B. Food budget C. Food habits D. dieting

50. The science that deals with the process of prevention of decay or spoilage of food thus allowing it to be stored in a fit condition for future use is

A. food pattern B. Food budget C. Food habits D. food preservation

SECTION B (40 marks)

ANSWER THE FOLLOWING TRUE OR FALSE

51. Functional foods provide health benefits beyond their nutrient contributions

52. Nutrient density is a measure of the energy a food provides relative to the amount of food (kcal per gram)

53. Nutrient dense food supply many nutrients that are critical to good health in comparison to their low-to-moderate number of kilocalories

54. Energy dense foods are foods that provide few/no nutrients for the number of kilocalories they contain.

55. Carbohydrate, Protein, Vitamins, Minerals, contain 13 – 36% of body nutrients

56. Organic literally means “dead”

57. Micronutrients are nutrients needed in small amounts by the body

58. Sugars, Starch and Fibers function as the body’s primary source of energy

59. Vitamins are essential for health in Cellular membrane structure and function, Myelin sheath in nervous system

60. Vitamins are No caloric, organic nutrients
61. Deficiencies of fat soluble vitamins may develop quickly if inadequate intake occurs
62. Fat soluble vitamins are stored in liver and fat tissue for long periods of time
63. Selenium sources include Brazil nuts, tuna, beef, and brown rice
64. Water is an Inorganic nutrient
65. Balance food is enough but not too much
66. Estimated Energy Requirement (EER) is dietary energy intake measured in kilocalories that is needed to maintain energy balance in a healthy adult.
67. Organic farms must be inspected monthly
68. The energy that runs most biological Systems on earth comes from solar energy
69. The Caloric needs of most organisms is measured as Basal Metabolic Rate (BMR)
70. Exercise produces a decrease in metabolic rate
71. < 20 or $> 30^{\circ}\text{C}$, metabolic rate is increased
72. Taller individuals have a higher BMR compared to shorter individuals.
73. Males tend to have higher basal metabolism than females
74. As we get older basal metabolic rate becomes increasingly slower
75. Anabolism is the process of building large molecules from simple molecules.
76. More than 80% of people with type 2 diabetes are overweight
77. Non-enzymatic reactions in food such as oxidation and mechanical damage can cause food spoilage
78. One reason for preserving food is for easy transportation
79. Preservation also aims to retain as many of the qualities of food
80. The term protein energy malnutrition was adopted by WHO in 1900
81. Stunting is insufficient weight gain relative to height
82. Kwashiorkor means sickness of weaning
83. Occasional signs of kwashiorkor is oedema
84. The term marasmus is derived from the Greek marasmus, which means stunting
85. Marasmus represents an adaptive response to starvation, whereas kwashiorkor represents a maladaptive response to starvation
86. Polyuria and polydipsia are clinical signs of hyperglycaemia
87. Prolonged duration or increased intensity of exercise can cause hyperglycaemia

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88. 45-65% kcal from protein is ideal
89. Gastric bypass can be employed in treating an overweight
90. The cause of primary hypertension is unknown

SECTION C

ANSWER USING UNITS PROVIDED IN THE PASSAGE (10 marks)

Dr Kyei weighs 102kg with a height of 1.95 meters

- A. Calculate his BMI indicating the formulae for the calculation of BMI (4)
- B. Indicate whether Dr. Kyei is overweight, underweight, moderately obese or morbidly obese (2)
- C. Indicate whether Dr. Kyei needs to lose weight or gain weight in order to obtain ideal weight (2)
- D. Calculate how many kilograms of weight Dr Kyei needs to be loose or gain in order to obtain ideal weight (2)

Good luck